MILITARY SPECIFICATION

LAUNCHER, GUIDED MISSILE AERO 5A-1

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 *Scope* - This Specification covers the requirements for the Aero 5A-1 Guided Missile Launcher used to retain and release the AGM-12B or AGM-45A missiles.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on date of i-on for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-corners

Bureau of Naval Weapons

WR-53	Naval. weapons Requirements Palletizing Domestic Unit Loads
WR-54	Naval Weapons Requirements Palletizing Fleet Issue Unit Loads

Military

MIL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture-and-Fungus-Resistant, of Comrnudcatd.ens, Electronic, and Associated Electrical Equipment
MIL-C-5015	Connectors, Electrical, "A" Type
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General. Specification for
MIL-H-6088	Heat Treatment of Aluminum Alloys

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MIL-H-6875	Heat Treatment of Steels (Aircraft Practices Process for)
MIL-G-7118	Grease, Aircraft Gear and Actuator Screws for Low and High Temperatures
MIL-F-7179	Finishes and Coatings, General Specification for Protection of Aircraft and Aircraft Parts
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-T-7743	Testing, Store Suspension Equipments General Specification for
MIL-A-8591	Airborne *Wines* and Associated Suspension Equipment, General Design Criteria for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-L-22769	Launcher, Weapons, Airborne and Associated equipment General Specification for
MIL-c-26482	Connectors, Electric, Circular, Minature, Quick Disconnect

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specification and Standards, Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

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DRAWINGS

Bureau of Naval Weapons

67A298	BuWeps Data List, Guided Missile Launcher, Aero 5A-1

2.2 Availability of Documents - When requesting specifications, standards, and drawings, refer to both title and number. Copies of this specification and other-applicable documents required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, U. S. Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence *over any* specification.

(b) This Specification - This specification shall have precedence over all applicable subsidiary specifications and drawings. Any deviation from this specification, or from subsidiary specifications, where applicable, shall be specifically approved in writing by the procuring agency.

(c) Referenced Specifications -Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall. apply to the extent specified.

3. REQUIREMENTS

301 Data - No data is required by this specification (other than reports accompanying samples submitted for testing), *or* by applicable documents referenced in 2.1 unless specified in the contract or order.

3.2 Parts and Materials - Parts and materials used in the manufacture of the launcher shall be in accordance with MIL-L-22769.

3.2.1 Selection - Specifications and standards *for* **all. parts** and materials which are not specifically designated herein shall be selected in accordance with MIL-STD-143.

3.2,2 Standard Parts - Standard parts shall be in accordance with MIL-L-22769.

3.2.3 Threaded Parts - Except for purchased parts with tapped holes, machine screw threads shall conform to Specification MIL-S-7742.

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3.2.4 *Fungus* - The Launcher shall be fungus-proofed in accordance with MIL-L-G22769.

3.3 Desire and Construction - The launcher shall be designed and constructed in accordance with drawings listed under DL 67A298 and this specification.

3.3.1 Conformance to Drawings - No departure from, revision or change to the drawings called out herein shall be made without prior approval of the procuring activity. The contractor shall notify the procuring activity of any conflict between engineering drawings and this specification.

3.3.2 Bomb Rack Compatibility - The launcher shall be capable of being directly attached to standard bomb racks utilizing 14-inch or 30-inch hook spacing and shall have (2) each 30" launch lugs-P/N 54A46c358 and 2 each 14" launch lugs P/N 1252628 packed with each launcher.

3.3.3 Interchangeability - Shall be in accordance with MIL-L-22769,

3.3.4 Size - Overall size of the launcher shall not exceed **75 inches** in length, 12 inches in width, and 8 inches in height, excluding launch lugs and tracks.

3.3.5 Weight - The weight of the launcher shall not exceed 110 pounds.

3.3.6 Finish - The launcher and its parts shall be finished in accordance with the applicable requirements of Specification MIL-F-7179. The exterior color shall be insignia white, color 17875, of FED-STD-595.

3.3.7 Heat Treatment - Heat treatment of aluminum alloys shall be in accordance with Specification MIL-H-6088. Steel parts shall be heat treated as specified in Specification MIL-H-6875.

3.3.8 Lubricants - All moving parts within the launcher shall be lubricated with a low temperature grease conforming to Specification MIL-G-7118, except trach bearing surfaces which shall. be coated with a commercial molybdenum disalfide lubricant such as Bel Ray Kote A2, or approved equivalent.

3.3.9 Transformers - All transformers shall be in accordance with the applicable drawings listed under DL 67A298.

3.3.10 Relays - All relays shall be in accordance with the applicable drawings listed under DL 67A295.

3.3.11 Solenoids - All solenoids shall be in accordance with applicable drawings listed under DL 67A298.

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3.3.12 Electrical Wiring - Electrical wiring shall meet the requirements of Specification MIL-W-5088.

3.3.13 Dissimilar Metals - Dissimilar metals shall be in accordance with MIL-L-22769 except the term "avoid" does not preclude the use of dissimilar metals provided they are not in direct contact with each other.

3.3.14 Nameplates and Product markings - A nameplate conforming with Specification MIL-N-18307 shall be attached securely to a fixed part of the launching equipment. Parts shall be marked in accordance with Standard MIL-STD-130. Nomenclature assignment, nameplate approval, and serial number assignment shall be in accordance with Specification MIL-N-18307.

3.4 Electrical Requirements -

3.401 Input Power - The launcher shall perform as specified herein when supplied with standard aircraft power meeting the requirements of MIL-STD-704, and d.c. voltage received from the AGM-12B missile battery.

Kind of power:

(1) 115 volts a.c. single phase; 400 cps; 60va, maximum, in accordance with Category "C" of MIL-STD-704.

(2) 28volts d.c.; 60va, amps maximum, in accordance with Category "C" of MIL-STD-704.

(3) upto 135 volts d.c.; 0.075 ampere Maximum (input from AGM-12B missile guidance and control section battery). The current is drawn by the launcher only when the ordnance firing relay is energized (see 3.4.4).

NOTE : When the Launcher is installed on an aircraft which cannot supply the required d.c. power (such as the Navy A4), the self-contained power supply specified in 3.4.2 must be installed and utilized. "When the power supply is used, the demand for aircraft a.c. power increases to:

115 volts a.c. single phase; 400 c.p.s; 860 va. maximum, in accordance with Category "C" of MIL-STD-704.

3.4.2 Self-contained Power Supply - The launcher shall contain a removable electrical power supply which shall convert the 115 volts a.c., 400 cps aircraft power specified in 3.4.1 to 18 to 31 volts d.c. and 28 plus or minus 3 volts a.c. at 400 CpS (nominal). The capability of the launcher to meet the structural design requirements herein shall not be affected whether the electrical components of this power supply are removed or installed. The 18-31. volts d.c. shall be utilized within the launcher to activate the missile retention mechanism (see 3.4.7).

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The 28 volts a.c. shall be routed to the crystal oven in the AGM-12B missile receiver during prelaunch warmup. The electrical components of this power supply must be physically removed from the casting upon which they are mounted or be electrically disconnected when the launcher is installed on aircraft which supply all required a.c. and d.c. power (see 3.4.1).

3.4.3 Filament Transformer - The launcher shall contain a filament transformer which shall convert 115 volts a.c. aircraft power into 7.5 volts a.c.. (nominal) filament voltage for use in the guidance and control section of the AGM-12B missile during prelaunch warmup.

3.4.4 Ordnance Firing Relay - The launcher shall contain an "ordnance firing relay" which shall delay initiation of the AGM-12B missile rocket engine at time of launch until the battery in the guidance and control section of the missile has built Up sufficient B+ voltage to insure missile response to early commands. This relay also operates a safety circuit to preclude inadvertent fuze function selection prior to initiation of missile launch.

3.4.5 Beacon ElectricalPackage - When required, the launcher shall contain a removable beacon electrical, package to provide 5.8 volts AC relaunch warmup current for the AN/DRT-3 Radio Beacon installed in the AGM-12-B Missile. The beacon package shall be capable of actuation by a launch signal from the aircraft as low as 18 VDC. The beacon electrical package may be optionally removed or installed in the launcher as needed.

3.4.6 Missile Launch Signal - The launcher shall provide an Immediate path for a 28 volt d.c. launch signal from the aircraft to either the AGM-12B or AGM-45A missiles through the energized retention mechanism relay contacts when the pilot depresses the "launch" switch. The launcher shall provide a delayed secondary path for this same voltage to the AGM-12B missile through the ordnance firing relay (see 3.4.4).

3.4.7 Missile Retention Mechanism Electrical Functions - The missile retention mechanism shall provide the following electrical safety features to prevent inadvertent launching or disengagement of the AGM-12B *or* AGM-45A missiles prior to the time the pilot energizes the armament buss in the aircraft.

(1) The missile shall be physically restrained from-leaving the launcher until the solenoid in the retention mechanism is energized,

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(2) An electrical interlock shall be provided for the launching circuits which shall preclude premature launching from inadvertent depressing of the launch switch in the aircraft or from extraneous voltages.

(3) Should the input voltage to the solenoid drop as low as 18 volts d.c. under emergency conditions the solenoid shall still energize.

3.4.8 Fuze Function Voltage - The launcher shall provide an electrical path from the aircraft to the AGM-12B missile for the fuze function voltage when the ordnance firing relay is energized (see 3.4.4).

3.4.9 dielectric strength - The dielectric shall be capable of withstanding, without failure, an electromotive force of 1056 volts a.c. at 50 to 100 cps for one minute, applied between each isolated electrical circuit and all other isolated electrical circuits, and between each isolated circuit and structural ground.

3.4.10 Insulation Resistance - The insulation resistance of the electrical circuits and devices shall be 100 megohms or more when measured at a constant potential of 500 volts d.c. by means of a megohmeter or other suitable method between each isolated circuit and all. other isolated circuits and between each isolated circuit and structural ground.

3.4.11 Electrical Continuity - Electrical continuity not exceeding 0.3 ohm shall exist in all complete electrical circuits of the launcher. including switch and relay contacts, but excluding solenoid and transformer inductive windings and other electrical components.

3.4.12 Electrical Connectors - Electrical connectors shall conform to Specifications MIL-C-5015 and MIL-C-26482, as applicable,

3.5 Service Conditions - The launcher shall function as specified herein under any or all of the following environmental conditions:

3.5.1 Temperature - The launcher shall be capable of withstanding in the operating and nonoperating condition exposure to temperatures within the range of -65°F to 160°F . Temperatures may remain constant up to 160°F and may be held for five minutes up to 160° F. Temperatures may vary at a rate as high as 1°F per second.

3.5.2 Altitude - In accordance with MIL-L-22769 except delete 70,000 feet and substitute 50,000 feet.

3.5.3 Salt Spray - In accordance with MIL-L-22769.

3.5.4 Humidity - In accordance with MIL-L-22769.

3.5.5 Sand and Dust - In accordance with MIL-L-22769.

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3.5.6 Missile Retention Mechanism - The launcher shall contain a forward retention mechanism. This mechanism consists of a ratio linkage and a solenoid. When the mechanism is Unlocked, a compression spring connected to the linkage restrains the missile with a force of no less than 837 pounds and no more than 2079 pounds. When the mechanism is locked, the missile is restrained from leaving the launcher and self contained switches prevent ignition of the missile's rocket engine.

3.5.7 Vibration - The launcher shall be capable of meeting the vibration environment imposed by the test procedures of Specification MIL-T-7743, except that the support points shall be considered to be the attachment points of the missile to the launcher and the maximum frequency of vibration shall be 300 cps and the destructive vibration shall have a total duration of 1.5 hours each for both the loaded and unloaded conditions.

3.5.8 Structural Strength - The launcher shall be capable of withstanding the critical forces specified in Specification MIL-A-8591 for wing mounted stores of 500 pounds or less up to design limit load, shall not exhibit permanent deformation detrimental to operation when subjected to 115 percent of design limit load (yield load), and shall not prematurely release the missile when subjected to 150percent of design limit load (ultimate load).

3.5.9 Shock - The launcher, with missile installed, shall be capable of withstanding dynamic, loads related to the critical flight, catapulting, and arrested landing conditions specified in Specification MIL-A-8591 for wing mounted stores of 500 pounds or less. The launcher, without missile installed, shall be capable of withstanding a dynamic load of 25 g along each of the principal axes.

3.6 Reliability - The launcher shall be capable of meeting the following reliability requirements. Acceptable demonstration testing of these requirements during the preproduction testing program shall serve as certification of contractor compliance therewith.

3.6.1 Operational Stability - The launcher shall be capable of 100 actual or simulated launchings without the necessity of servicing or replacement of parts.

3.6.2 Total Operating Life - The launcher shall be capable of launching 500 missiles with reasonable servicing and replacement of parts. Acceptable demonstration testing of this requirement during the preproduction tests (see 4.2) shall serve as certification of contractor compliance.

3.7 Maintainability - The launcher shall be capable of disassembly and replacement of parts without the use of special tools,

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or

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any commercial laboratory acceptable to the procuring activity. The procuring activity reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

40101 Classification of Tests - The launcher shall be subjected to the following tests to determine compliance with the requirements of this specification:

(1) Preproduction teats
(2) Acceptance tests

4.1.2 Test Samples - The launchers used for preproduction and sampling tests shall be supplied over and beyond the, quantity of equipments specified for delivery in the contract. At the conclusion of testing samples may be returned to the contractor or retained, at the option of the procuring activity.

4.2 Preproduction Tests - Preproduction tests shall be conducted by the procuring activity On three (3) sample launcher furnished by the contractor for the purpose of Verifying that launchers produced by the contractors production methods and tools meet all requirements of this specication. The first three launchers produced by the contractor shall. be shipped at the contractors expense to a testing facility designated by the procuring activity for this purpose. The sample launchers shall be completely representative of the launchers to be delivered under the contract and shall have successfully passed the individual tests specified in 4.3.1 prior to shipment. A copy of the individual tests results certified by the cognizant Government inspector shall be forwarded with the samples. Any defects or failures occurring during the Government tests shall be cause for tentative rejection of the preproduction samples. All rejected samples may be returned to the contractor for rework or correction of defects when considered practical and feasible, otherwise the contractor will be required to submit additional samples for retest in lieu of reworked samples. Preproduction tests shall consist of all tests listed under 4.5.

4.2.1 PreProduction Approval - Approval of the preproduction samples shall be by the procuring agency upon satisfactory completion of all tests~ No production equipments shall be delivered prior to approval of the preproduction samples, except when conditional acceptance is specifically authorized by the Contracting Officer. Prefabrication of production equipment and shipment of conditionally accepted units prior to the approval of the preproduction models is at the contractor's own risk. The approved preproduction samples may be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance.

4.3 Acceptance Tests - Acceptance of equipment shall be on a lot basis. Lots shall consist of launchers which have passed individual

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tests. Acceptance, tests shall be conducted during the course of the contract to assure that the launchers produced continue to meet the requirements of this specification. Acceptance tests shall consist of the following:

(1) Individual tests
" (2) Sampling tests

4.3.1 Individual Tests - Individual tests shall be accomplished, under the responsibility of the contractor, on each equipment submitted for acceptance under the contract. The contractor shall-maintain test reports showing the quantitative results for all individual tests, Such reports shall be signed by an authorized representative of the contractor and this data shall be made available to the procuring agency upon request. Individual tests shall consist of the following tests:

(1) Examination of product
(2) Operational adequacy
(3) Dielectric strength
(4) Insulation resistance
(5) Electrical continuity

4.3.2- Sampling tests shall be conducted at a Government designated la oratory testing facility under the responsibility of the procuring activity on launchers selected at random by the Government inspector from production lots offered for acceptance. The purpose of these tests till be to assure the continual conformance of production launchers with the preproduction test criteria. Launchers selected for those tests shell be forwarded at the contractors expense to the Government selected laboratory for testing, and shall have previously passed the individual tests specified in 4.3.1. Samples shall be suitably labeled to identify them with the particular production lot from which they were selected. Sampling tests shall consist of all tests listed under test methods (4.5).

4.3.2.1 Sampling Test Schedule - Unless otherwise specified in the contract, the number of sample launchers required to be tested shall be as follows:

Size of Production Lot Offered for Acceptance, (see 4.3.2.2)	Quantity of Samples Required
99 or less	2
100 to 299	3
300 to 499	4
500 or more	4 plus 1 for each additional 500 Or fraction thereof

4.3.2.2 Definition of Production Lots -A production lot shall be defined as a quantity of launchers having the same nomenclature and part number produced by the same manufacturer in continuous productions and offered for acceptance as one of the production lots required for shipment by the contract.

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4.3.2.3 Lot Size - Unless otherwise specified by the procuring activity, the size of each production lot shall be the quantity of launchers required for delivery each month. Production lots shall include the necessary extra launchers required for sampling tests (see 4.1.2). The extra launchers shall not be considered in determining the lot size.

4.3.2.4 Approval of Production Lots - A production lot shall receive final approval from the procuring activity after satisfactory completion of the sampling tests conducted on the lot samples.

4.4 Test Conditions - Unless otherwise specified, the launcher shall be subjected to tests under the following conditions:

(1) Temperature:	Ambient 65 degrees F to 95 degrees F
(2) Altitude:	Normal ground elevation
(3) Vibration	None
(4) Humidity	Ambient to 95 percent

4.4.1 Tolerances - Unless otherwise specified, the maximum allowable tolerance on test condition measurements are as follows:

(1) Temperature:	Plus or minus 5 degrees F
(2) Altitude:	Plus or minus 5 percent in feet
(3) Humidity:	Plus or minus 5 percent relative
(4) Vibration Frequency	Plus or minus 2 percent
(5) Vibration Amplitude:	Plus or minus 5 percent

4.4.2 Special Test Equipment - The supplier shall furnish and maintain all special test equipment or special instruments required in performing the individual tests required under paragraph 4.3.1 herein. Unless values are given for individual tests, all special test equipment employed as a means for testing the launcher shall. be of such nature and design that any errors produced to not exceed 25 percent of the maximum allowable tolerance established for the value to be measured.

4*5 Test Methods -

4.5.1 Examination of Product - Each launcher shall be examined, carefully to determine that no defects immaterial or workmanship exist which would adversely affect performance.

4.5.2 Operational Adequacy - operational adequacy shall consist of the following mechanical and electrical tests:

(1) Mechanical tests shall consist of a functional check of all mechanical components to demonstrate that they operate freely without sticking or binding.

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(2) Electrical tests shall consist of an electrical continuity check, a check on transformer output and relay operation, and actuation of the retention mechanisms

NOTE: These tests shall not include simulated launch or captive flight loads.

4.5.3 Dielectric Strength - An electromotive force of 1056 volts a.c. at 50 to 100 cps shall be applied for one minute between each isolated electrical circuit and between each isolated circuit and structural ground. Any breakdown, arcing, or steady increase of fluctuation in leakage current shall constitute a failure.

4.5.4 Insulation Resistance - The insulation resistance of the circuits and components Shall be measured at a constant potential of 500 Volts d.c. by means of a megohmeter or other suitable method. The potential shall be applied between each isolated circuit and between each isolated circuit and structural ground for a period of approximately one minute. A measured insulation resistance of less than 100 megohms shall be cause for rejection.

4.5.5 Electrical Continuity - The electrical continuity of each complete electrical circuit in the launcher shall be measured, The resistance shall not exceed 0.3 ohm, including all relay and switch contacts, but excluding transformer and solenoid inductive wingings and other electrical components.

4.5.6 Environmental Tests: Launchers submitted as samples in accordance with 4.2 and 4.3.2 shall be subjected to the environmental tests outlined below,

4.5.6.1 High Temperature - The high temperature test shall be conducted in accordance with procedure 11 of Specification MIL-E-5272. At the conclusion of the 48-hour exposure to the 160°F environment the operational adequacy test specified in 4.5.2 here-in shall be performed.

4.5.6.2 Low Temperature - The temperature test shall be conducted in accordance with procedure I of Specification MIL-E-5272, except that the test for performance under icing conditions of Specification MIL-T-7743 shall be included.

4.5.6.3 Humidity - The humidity test shall be conducted in accordance with precedure I of Specification MIL-E-5272 with all electrical connectors and receptacles protected by mating connectors, except that in

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addition to the functional test after exposure, one functional electrical test shall be conducted during the condition of 160°F and 95 percent humidity ,

4.5.6.4 Altitude- The altitude test shall be conducted in accordance with Specification MIL-E--5272, procedure VI, condition C.

4.5.6.5 Salt Spray - The salt spray test shall be conducted in accordance with procedure I of Specification MIL-E-5272, using a five percent salt solution. Exposure time shall be 50 hours. All electrical connections shall be suitably protected by the mating connector for this test. Items undergoing this test shall be inspected and operated within one hour after completing of the test- Following wash down in accordance with MIL-E-5272 the allowable dry time prior to operational tests shall be one hour.

4.5.6.6 Sand and Dust - Sand and dust tests shall be conducted in accordance with procedure I of specification *MIL-T-77432 including* the frozen sand test.

4.5.6.7 Vibration - Vibration tests shall be conducted as specified in Specification MIL-T-7743, except that the SUPPOrt POints shall be Considered to be the attachment points of the missile to the launcher, and that maximum frequency of vibration shall be 300 cps and destructive vibration shall have 8 total duration of 7.5 hours each for both the loaded and unloaded conditions-

4.5.6.8 ShOCk- Dynamic load tests of the launcher shall be conducted to the critical flight, catapulting, and arrested landing design conditions of Specification MIL-A-8591, as applied to a weight category of 500 pounds or less for wing mounted stores. Rotational accelerations may be omitted if not within the dynamic capability of the dynamic test equipment, provided the increases in reaction loads resulting from applied aerodynamic or inertia moments are accounted for by increasing the translational accelerations of the dynamic tests. The launcher shall be supporting the Bullpup missile (or simulated missile having equivalent mass and stiffness distribution). The maximum value of applied acceleration as measured at the center of gravity of the store, shall be attained in not more than 1.0 second and not less than 0.03 second after moment of impact. The dynamic tests shall be conducted according to the following schedule:

(a) The launcher shall be tested twice at loads between 1.04 and 1.10 times the limit load condition. The launcher shall be inspected and shall operate in the prescribed manner as specified herein after each test. There shall be no deformation noticeable upon inspection.

(b) In order to check for no-load dynamic effects, the launcher shall be subjected to an acceleration of 25 gravitational units along all the principal axes while supporting no-load. Accelerations shall be measured on the *launcher* for these tests.

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4.5.6.9 Static Load - Static load tests of the launcher shall be conducted with a Bullpup missile (or simulated missile having equivalent sttifness distribution) attached as required for a service installation. The launcher shall be attached to a bomb rack that is bolted to a structure which approximates the attachment of the bomb rack to the aircraft. The launcher shall be tested to each design limit condition for flight loads as shown on Figure 1. Following each test limit loading, the launcher shall operate normally, both electrically and mechanically, at room temperature and rated voltage. No deformation detrimental to the operation of the launcher is acceptable. Ultimate tests shall be conducted for each critical design flight condition in such a manner that the values of resultant load are increased in increment not greater than 10 percent of limit load until the design ultimate loading conditions are attained.

" **4.5.6.10** Operational Tests - Tests necessary to prove the operational capability shall be with a complete launcher mounted on a bomb rack to simulate a typical aircraft. installation. A dummy missile shall be suspended from the launcher, and loads applied to the missile to simulate aerodynamic thrust and applied "g" loads in accordance with Figure 2. A simulated launch shall be performed at the conclusion of this test.

4.5.6.11 Input Voltage - The launcher shall be tested to verify that satisfactory operation is obtained under the input voltage limits specified in 3.4.1 herein.

4.5.7 Life Test - Five hundred simulated launchings of a missile or simulated missile by the launcher shall be made at room temperature and at rated voltage of the specified missile with launcher in horizontal flight attitude. The time interval between releases shall be sufficient to prevent overheating of electrical parts. Reasonable servicing and replacement of parts shall be permissible during the test.

4.6 Rejections and Retest -

406.1 If any sample fails to pass the tests specified herein, another random sampling of the same size will be submitted. Failure of the second sampling shall cause the entire lot to be rejected. Rejected lots may be reworked and resubmitted for retest upon approval by the procuring activity.

5. PREPARATION FOR DELIVERY

5*1 Preservation, Packaging. and Packing - Unless otherwise specified in the contract or order, preservation, packaging, and packing shall be in accordance with levels A, B, or C as defined herein, the applicable level to be specified in the contract or order (see 6.2).

5.1.1 Preservation and Packaging

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5.1.1.1 Levels A and B - If the contract or order specifies levels A or B, each launcher shall be preserved and packaged in accordance with method II of Specification MIL-P-116.

5.1.1.2 Level C - When level C is specified, each launcher shall be preserved and packaged in accordance with the manufacturer's commercial practice. It shall be the responsibility of the contractor to afford a level of protection adequate to prevent damage during transit from the contractor's plant to the first destination.

5.1.2 Packing

5.1.2.1 Level A - When level A is specified as the packing requirement, each launcher be packed in overseas containers conforming to Specification PPP-B-601 or PP-B-621. These containers should be palletized in accordance with the general requirements of WR-54 (minimum of nine containers in a palletized load).

5.1.2.2 Level B - When level B is specified as the packing requirement, each launch-l be packed in domestic containers conforming to Specification PPP-B-601 or PPP-B-621. These containers should be palletized in accordance with the general requirements of WR-53 (minimum of nine containers in a palletized load).

5.1.2.3 Level C - When level C packing is specified, each launcher shall be packed in containers acceptable by the common carrier which will insure safe delivery at the lowest rate to the point of delivery. Containers shall comply with the carrier's rules and regulations applicable to the mode of transportation.

5,2 Marking - All marking for interior and exterior containers shall conform to MIL-STD-129 and shall include the following:

(a) Launcher, Guided Missile Aero 5A-1

(b) Name of supplier (and manufacturer if not the same)

(c) Date of preservation

(d) Specification number

(e) Contract number

(f) Lot number

(g) Weight

6. NOTES

6.1 Intended Use - The Aero 5A-1 Guided Missile Launcher is intended to be suspended from standard aircraft bomb racks utilizing 14-inch or 30-inch hook spacing for the purpose of suspending, carrying, and launching Bullpup AGM-12B or Shrike AGM-45A guided missiles.

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6.2 Ordering Data - Procurement documents should specify the following:

(1) Title, number, and date of this specification

(2) Selection of applicable levels of preservation, packaging, and packing (see 5.1).

(3) The laboratory facilities to be utilized for conducting preproduction and production lot sample tests.

(4) Production sampling lot size

(5) Beacon electrical package requirements (see 3.4.5).

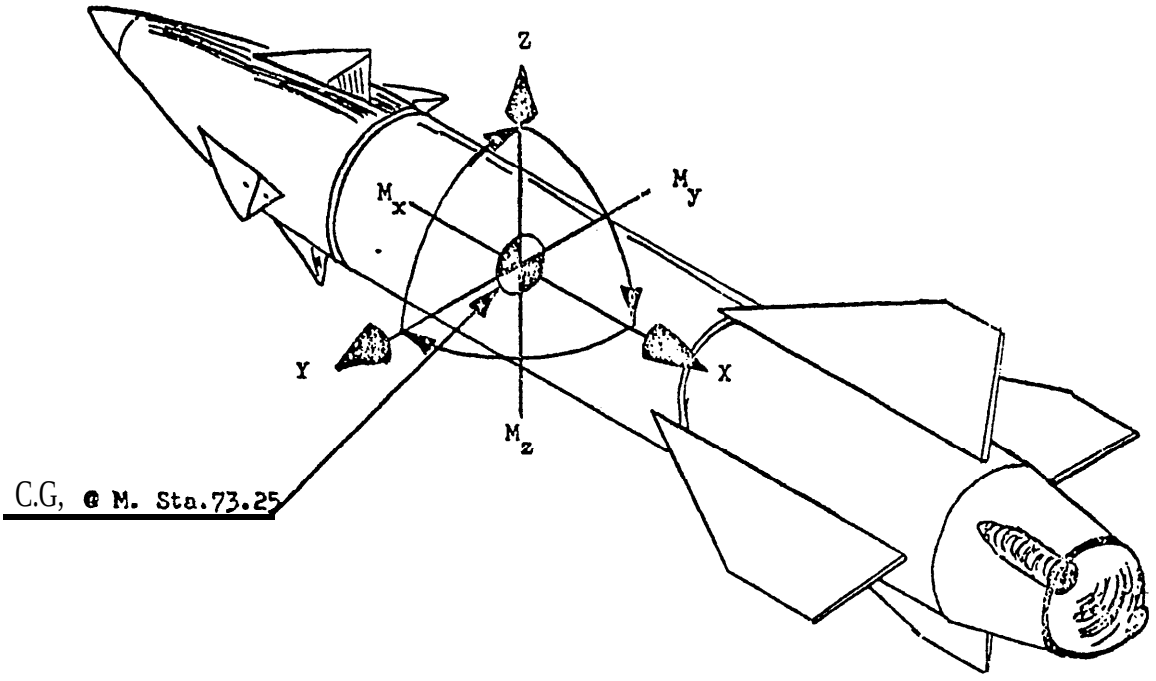
(6) A4 power supply requirements (see 3.4.2).

6.3 Definitions

6.3.1 Missile Designations - The AGM-12B missiles referred to herein were formerly designated as ASM-N-7a and GAM-83A (Bullpup) guided missiles. The AGM-45A missiles were formerly designated as ASM-N-10 (Shrike) missiles.

6.3.2 Alternating Current - All a.c. values expressed herein are in terms of root mean square.

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LOAD CONDITION (see MIL-A-8591	X	Y	Z	M_x	M_y	M_z
Condition 6	2527	6607	-5210	8640	-7941	5382
Condition 2	2416	1867	-8390	4800	-7941	2350
Catapult	5205	-		-	-	-

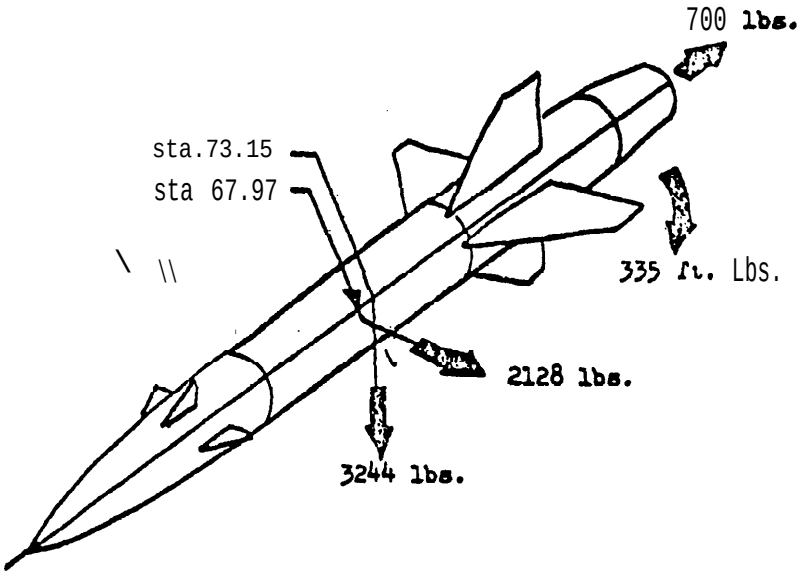
NOTES: 1. Installation on left wing.

2. Loads in pounds, moments in inch-pounds

30 Catapult condition is critical only for aft retention Apply drag load *only.*

FIGURE 1. AGM-12B CAPTIVE FLIGHT LIMIT LOADS

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LAUNCH LOADS EASED ON THE FOLLOWING CONDITIONS:

1. Altitude = Sea Level
2. Mach No. at Launch = 0.95
3. Missle Incidence to Airplane C =0 to -3°
4. Angle of Attack of the Airplane = -1°
5. Yaw Angle = ± 3°
6. Inertial Loads = 2.67g Vertical, l.Og Horizontal
7. Engine thrust not indicated

NOTES :

1. Inertial loads shown added to aerodynamic loads.
2. Loads are in body axis system.
3. Moments are taken about Station 66 at centerline.

FIGURE 2. *AGM-12H LAUNCH* LOADS-(LIMIT)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bereau No. 119-RO04

INSTRUCTIONS

This sheet is to be filed out by rsonnel either Government or contractor, involved in the use of the specification in procurement of products for ul timate use by the Department of Defense, This sheet is provided for obtaining information on the use of this specification which will Insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments form will be appreciated. Fold on lines on reverse side, staple in corner, and send to prepar'ng activity (as indicated on reverse hereof).

SPECIFICATION
MIL-L-81437(AS) LAUNCHER, GUIDED MISSILE AERO 5A-1

ORGANIZATION (of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

MATERIAL PROCURED UNDER A

U DIRECT GO VERNMENT CONTRACT m SUBCONTRACT

1, HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USEI

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMENDATIONS FOR CORRECTING THE DEFICIENCIES.

COMMENTS ON ANY SPECIFICATION RETIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

0 YES O NO If "YES", IN WHAT WAY?

REMARKS (Attach any pertinent data which may be of use in improving this specification, If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

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